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Park

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(54) **COSMETIC INJECTOR ON WHICH MICRONEEDLES FOR INDUCING ABSORPTION OF COSMETICS ARE MOUNTED, AND MANUFACTURING METHOD THEREFOR**

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A61M 37/00 (2006.01)

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CPC ***A45D 34/04*** (2013.01); ***A61M 37/0015*** (2013.01); *A61M 2037/0023* (2013.01); *A61M 2037/0061* (2013.01)

(58) **Field of Classification Search**
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See application file for complete search history.

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(57) **ABSTRACT**

A cosmetic injector includes an accommodation part for accommodating cosmetics; a head part having a flow path through which cosmetics flow; a microneedle part fixed to the head part, and inserted into the skin so as to cause the cosmetics to penetrate the skin; and a pumping part for transferring, to the head part, the cosmetics accommodated in the accommodation part. The head part includes: a first head part for receiving the cosmetics from the pumping part; and a second head part coupled to the upper portion of the first head part, and the microneedle part includes: a microneedle plate provided between the first head part and the second head part; and a plurality of microneedles which extend from the microneedle plate so as to protrude toward the upper surface of the second head part, and which guide, into the skin, the cosmetics discharged from the head part.

11 Claims, 11 Drawing Sheets

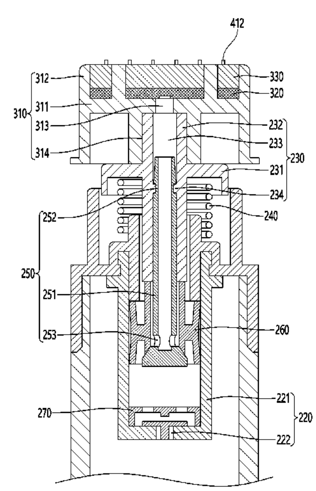


FIG. 1

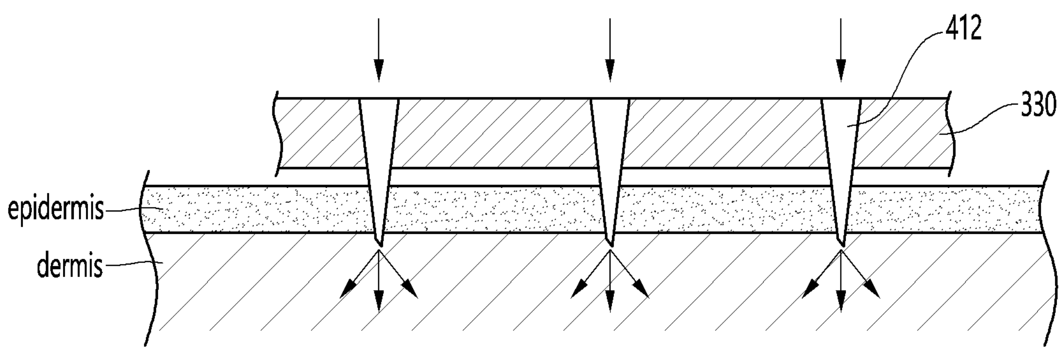
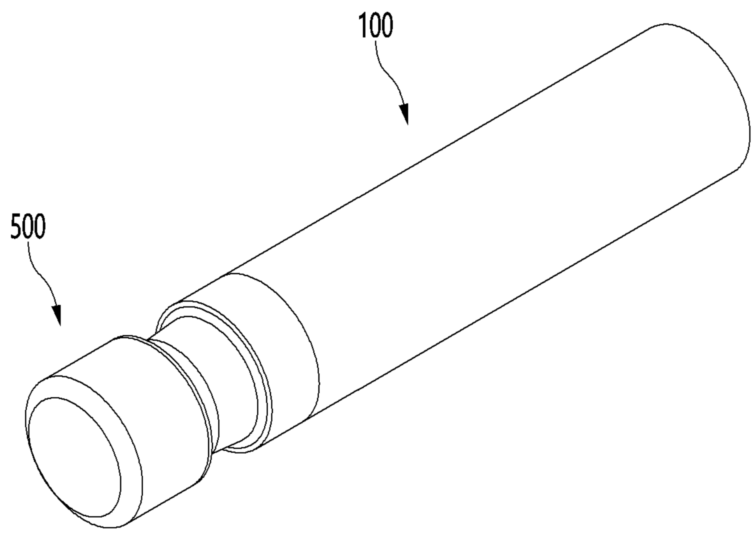
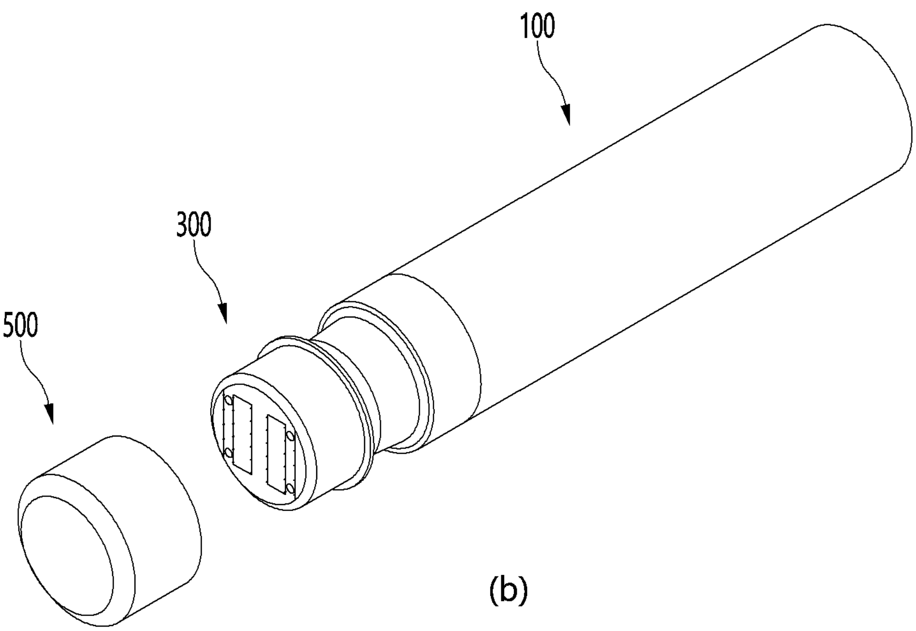


FIG. 2a



(a)

FIG. 2b



(b)

FIG. 3

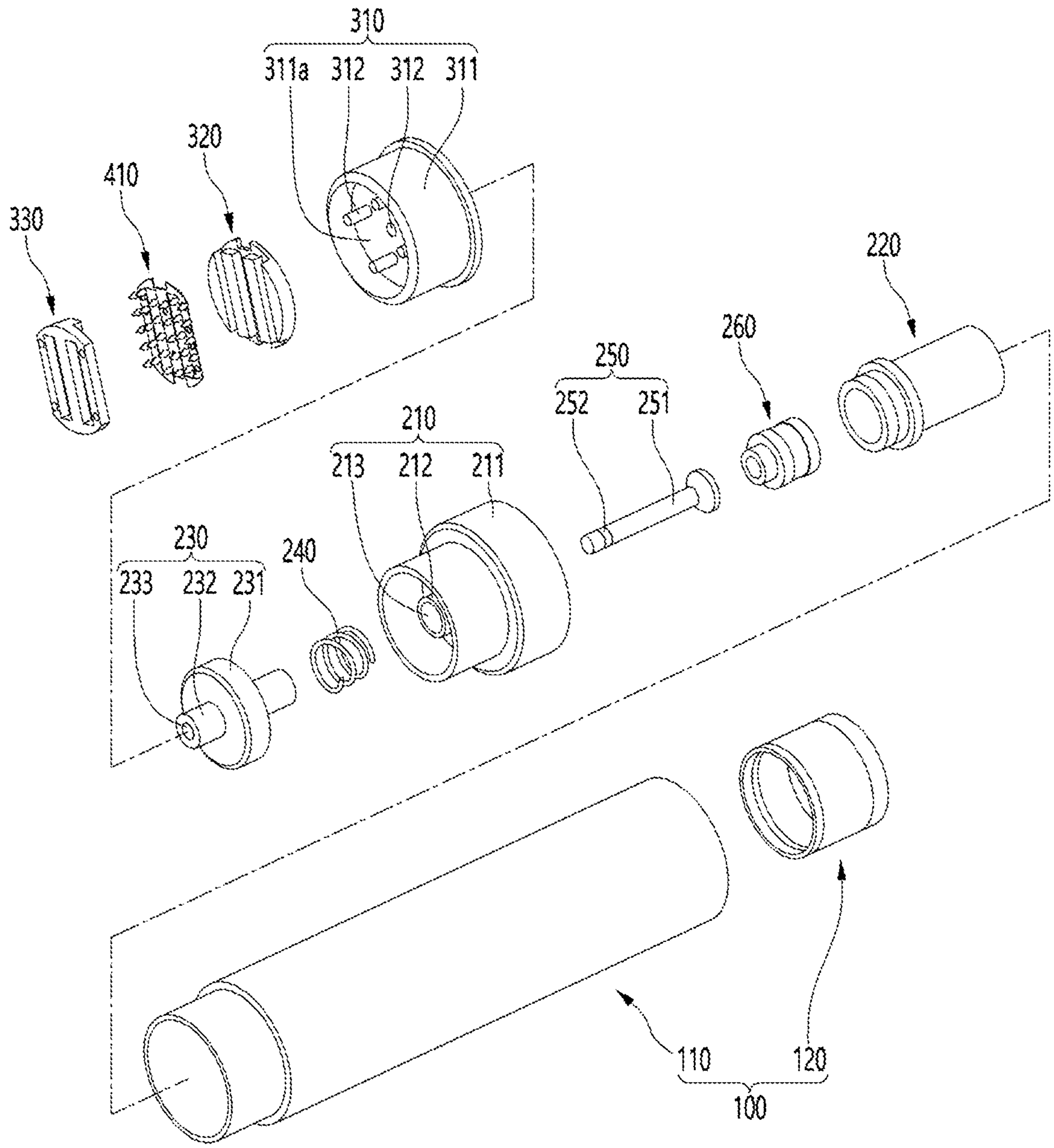

310
311a
312
312
311
320
410
330
220
260
250
252
251
210
213
212
211
230
233
232
231
240
110
120
100

Fig. 4

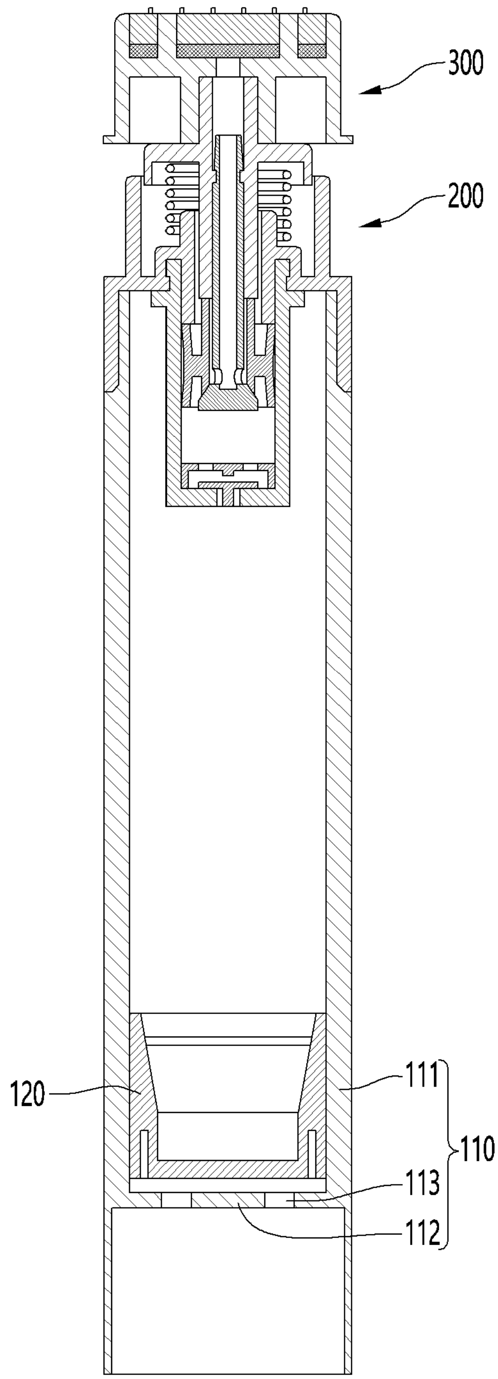


FIG. 5

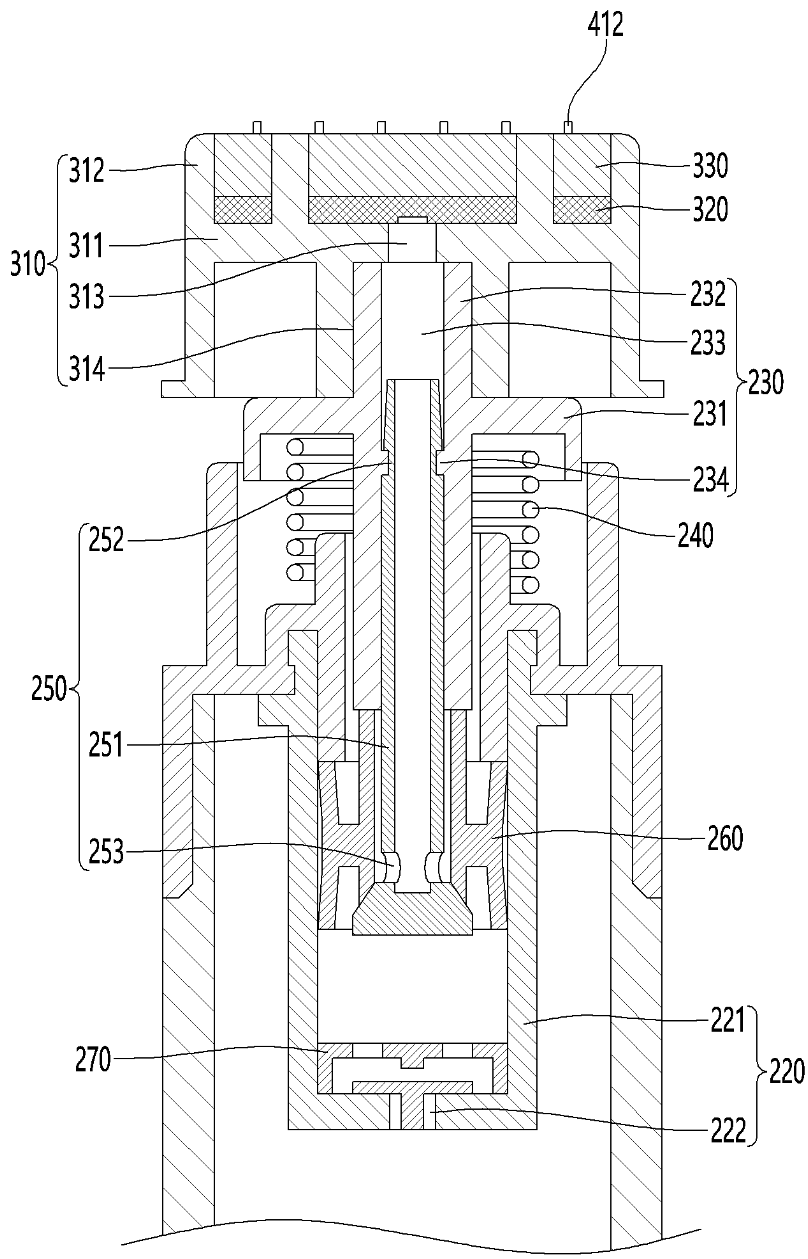

412
312
330
320
310
311
313
232
233
314
230
231
234
252
240
250
251
260
253
221
270
220
222

FIG. 6

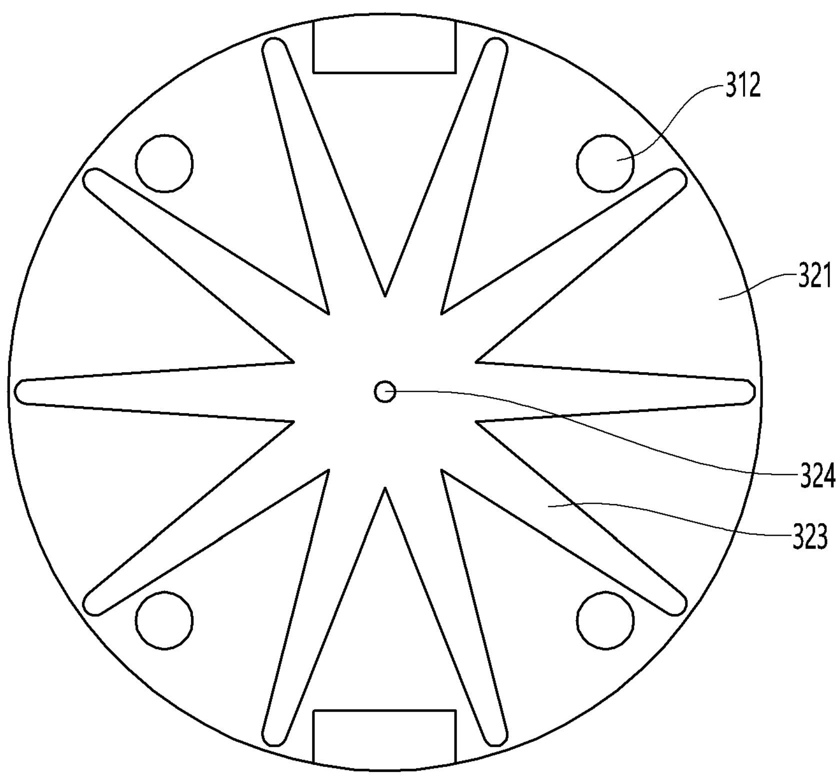


FIG. 7

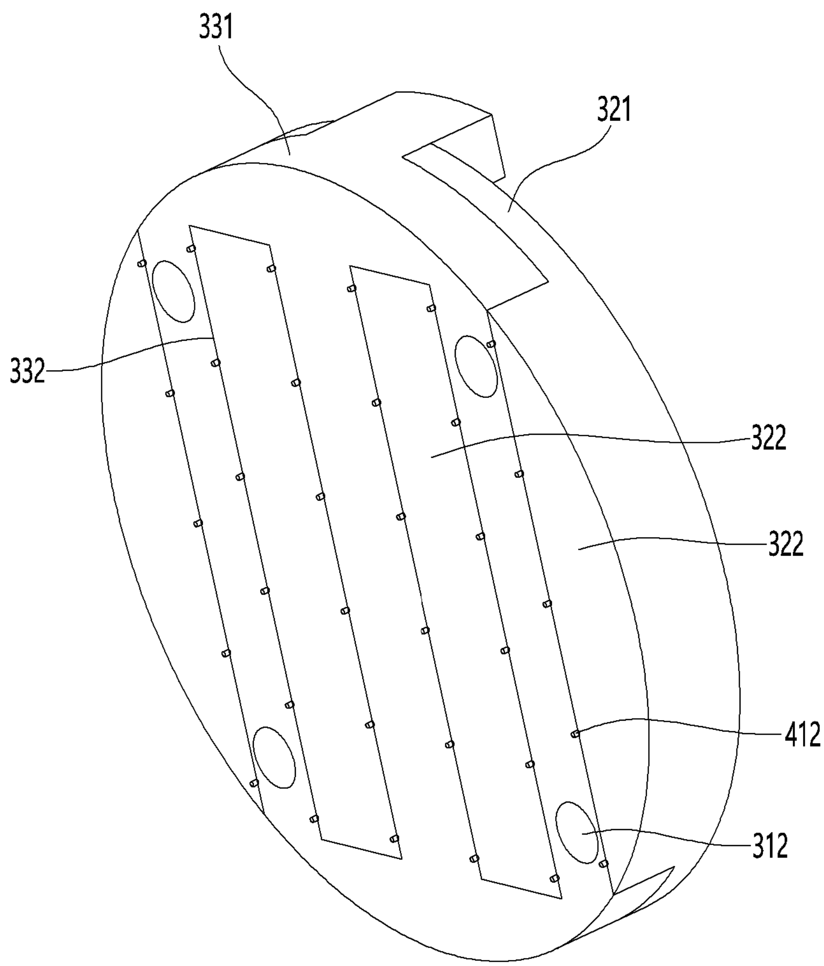


FIG. 8

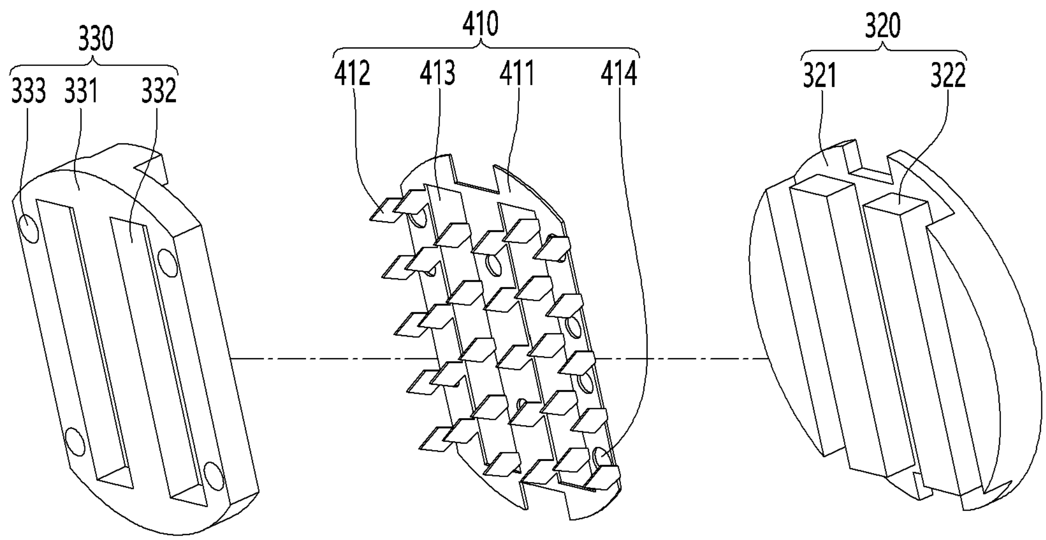


FIG. 9

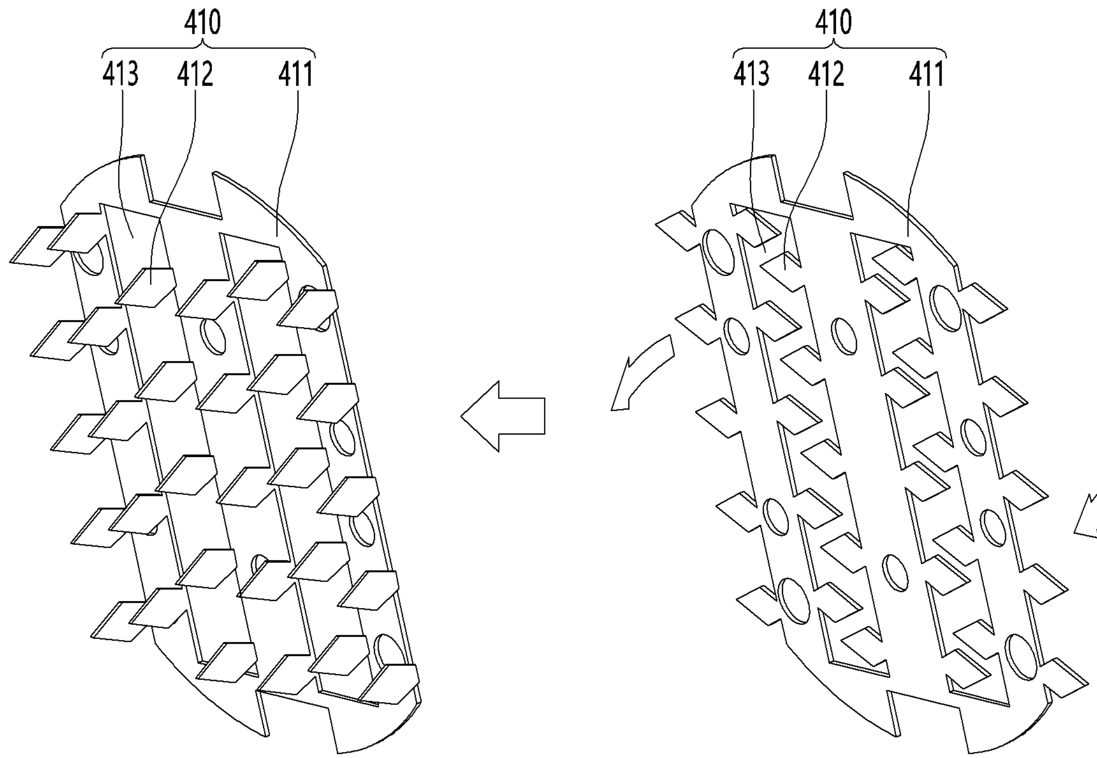
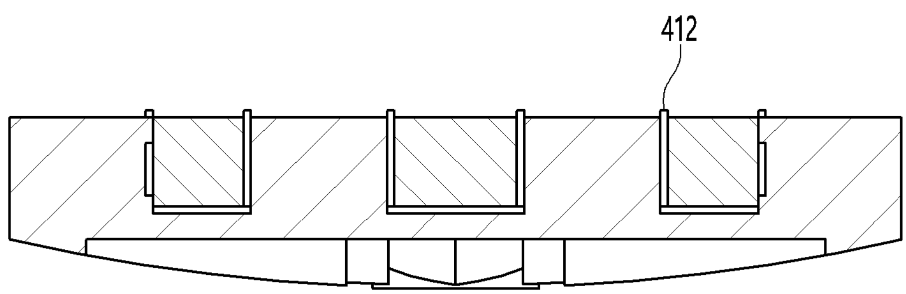


FIG. 10



COSMETIC INJECTOR ON WHICH MICRONEEDLES FOR INDUCING ABSORPTION OF COSMETICS ARE MOUNTED, AND MANUFACTURING METHOD THEREFOR

This application is a 371 of International Application No. based on PCT/KR2021/002641 filed on Mar. 4, 2021 which claims priority to Korean Application No. 10-2020-0041515, filed Apr. 6, 2020 and Korean Application No. 10-2020-0041529, filed Apr. 6, 2020, both of which are incorporated by reference, as if fully set forth herein.

TECHNICAL FIELD

The present invention relates to a cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, and a manufacturing method therefor.

BACKGROUND ART

Human skin is made of an epidermal layer and a dermal layer including collagen, and the amount of collagen in the dermal layer decreases as aging progresses. Accordingly, the skin becomes dry or wrinkled, which reduces the elasticity of the skin.

Therefore, nutrients such as vitamins or peptides that help collagen production are applied to the skin so as to keep the skin elastic. In general, the amount of nutrients, which are applied to the skin, pass through the epidermal layer, and are then absorbed into the dermal layer, is only 0.3% of the total content.

In order to solve the above limitation, a method of directly injecting nutrients into the subcutaneous tissue using a syringe is employed. This method also has several limitations, such as causing bleeding, pain, and inflammatory reactions, and leaving scars. Also, it may be difficult to apply this method depending on ages or skin characteristics.

As an alternative to this method, a method of injecting nutrients into the subcutaneous tissue by making micro-gaps in the skin using a microneedle is emerging.

This method can minimize pain, bleeding, and inflammatory reactions when injecting nutrients, and enables local injection of the nutrients, thus making it possible to effectively and continuously inject the nutrients only into desired areas. Therefore, this method is being actively studied.

The microneedle is usually a metal which has a tip made of micro units, and a cosmetic injector using a microneedle according to the related art has a structure in which a plurality of microneedles are individually mounted to a head part.

Therefore, there is a limitation in that each microneedle has to be mounted on the head part when manufacturing an instrument, and each microneedle can be broken and stuck in the skin while being inserted into the subcutaneous tissue and cause inflammatory reactions or other diseases.

DISCLOSURE OF THE INVENTION

Technical Problem

The purpose of the present invention is to provide a cosmetic injector in which a microneedle can be easily mounted on a head part.

Also, the purpose of the present invention is to prevent a microneedle mounted on a head part from being broken.

Also, the purpose of the present invention is to prevent a microneedle from being contaminated by an operator when a cosmetic injector is manufactured.

Technical Solution

According to an embodiment of the present invention, provided is a cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, the cosmetic injector including: an accommodation part configured to accommodate cosmetics; a head part having a flow path through which the cosmetics flow in; a microneedle part which is fixed to the head part, and which is inserted into the skin and allows the cosmetics to penetrate the skin; and a pumping part configured to transfer, to the head part, the cosmetics accommodated in the accommodation part, wherein the head part comprises: a first head part configured to receive the cosmetics from the pumping part; and a second head part coupled to the upper portion of the first head part, wherein the microneedle part comprises: a microneedle plate provided between the first head part and the second head part; and a plurality of microneedles which extend from the microneedle plate and protrude toward the upper surface of the second head part, and which guide, into the skin, the cosmetics discharged from the head part.

In an embodiment, the microneedle plate and the plurality of microneedles may be integrally formed.

In an embodiment, the first head part may include: a first head part body having the upper surface on which the microneedle plate rests; and a resting protrusion having the outer circumferential surface that contacts the plurality of microneedles.

In an embodiment, the resting protrusion may be provided in plurality.

In an embodiment, the microneedle plate has a resting protrusion through-hole through which the resting protrusion passes.

In an embodiment, the plurality of microneedles may extend upward from the vicinity of the resting protrusion through-holes of the microneedle plate or from edges of the microneedle plate.

In an embodiment, the microneedle part may be made of metal so that contact points between the microneedle plate and the plurality of microneedles are pressed and bent by the resting protrusion.

In an embodiment, the second head part may include a second head part body having the lower surface that contacts the microneedle plate, and the second head part body may have a resting protrusion coupling part that is a hole into which the resting protrusion and the plurality of microneedles are inserted.

In an embodiment, gaps configured to accommodate the plurality of microneedles may be formed between the resting protrusion coupling part and the resting protrusion or between the outer boundary of the second head part body and the resting protrusion.

In an embodiment, the length of each of the plurality of microneedles may be greater than the distance from the upper surface of the first head part body to the upper surface of the second head part.

In an embodiment, the cosmetic injector may further include a head housing that accommodates the first head part, the second head part, and the microneedle part, and receives the cosmetics from the pumping part.

In an embodiment, the first head part body may have: a guide hole that vertically passes through the first head part body; and a first head part flow path which is provided in the lower portion of the first head part body and formed radially about the guide hole, wherein the cosmetics, which have flowed into the head housing via the pumping part, are guided to the guide hole via the first head part flow path.

In an embodiment, the second head part may include

a second head part body having the lower surface that contacts the microneedle plate,

wherein the head housing further comprises:

a head housing body configured to accommodate the first head part, the second head part, and the microneedle part; and a fixing protrusion that protrudes from the head housing body, wherein the first head part further comprises a first head part fixing protrusion insertion hole through which the fixing protrusion passes, the microneedle part further comprises a microneedle part fixing protrusion insertion hole through which the fixing protrusion, which has passed through the first head part fixing protrusion insertion hole, passes, and the second head part further comprises a second head part fixing protrusion insertion hole through which the fixing protrusion, which has passed through the microneedle part fixing protrusion insertion hole, passes.

In an embodiment, the fixing protrusion may be provided in plurality, and the number of each of first head part fixing protrusion insertion holes, microneedle part fixing protrusion insertion holes, and second head part fixing protrusion insertion holes may correspond to the number of fixing protrusions.

In an embodiment, when the head part is pressed toward the pumping part, the cosmetics, which have been supplied via the pumping part, may be discharged via the outer circumferential surfaces of the plurality of microneedles.

According to an embodiment of the present invention, provided is a method for manufacturing the cosmetic injector of claim **1**, on which microneedles for inducing absorption of cosmetics are mounted, the method comprising: a resting operation of resting the microneedle plate on the upper surface of the first head part; and a head part coupling operation of coupling the second head part to the upper portion of the first head part.

In an embodiment, the microneedle part may be formed such that the microneedle plate and the plurality of microneedles are provided parallel to each other prior to performing the resting operation.

In an embodiment, the plurality of microneedles may be bent perpendicularly to the microneedle plate while the microneedle plate rests on the first head part.

In an embodiment, the plurality of microneedles may be bent simultaneously.

According to an embodiment of the present invention, provided is a microneedle part used in the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted.

Advantageous Effects

The present invention has the effect of providing a cosmetic injector in which a microneedle can be easily mounted on a head part.

Also, the present invention has the effect of preventing a microneedle mounted on a head part from being broken.

Also, the present invention has the effect of preventing a microneedle from being contaminated by an operator when a cosmetic injector is manufactured.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** shows a state in which cosmetics penetrate into the skin via a cosmetic injector.

FIG. **2**(*a*) is a perspective view of a cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention.

FIG. **2**(*b*) is a perspective view of a cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention and illustrates a state in which a cap part is open.

FIG. **3** is an exploded perspective view of the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention.

FIG. **4** is a cross-sectional view of the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention.

FIG. **5** is a cross-sectional view of a head part included in the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention.

FIG. **6** is a plan view of the head part included in the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention.

FIG. **7** is a perspective view of the head part included in the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention.

FIG. **8** is an exploded perspective view of the head part included in the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention.

FIG. **9** illustrates a process of bending microneedles of a microneedle plate included in the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention.

FIG. **10** illustrates the upper side of the head part of the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to the present invention.

MODE FOR CARRYING OUT THE INVENTION

Hereinafter, a cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, will be described with reference to drawings.

As used herein, the term of "microneedle" may refer to an entire needle made of metal, and a portion thereof inserted into the skin may have a length of 1 mm or less.

FIG. **1** shows a state in which a plurality of microneedles **412** are inserted in to the dermis after passing through the epidermis, and cosmetics including nutrients may reach the dermis via the surfaces of the plurality of microneedles **412**.

That is, when the microneedles **412** pierce the epidermis of the skin, the cosmetics may penetrate into a portion under the epidermis of the skin via the micropores which are formed in the epidermis of the skin by the microneedles **412**.

Referring to FIGS. **2**(*a*) and **2**(*b*) showing the perspective views of a cosmetic injector for absorbing cosmetics, the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to an embodiment of the present invention may include: an accommodation part **100** that accommodates cosmetics including nutrients; a head part **300** that has a flow path, through which the cosmetics flow in, and is in close contact with the skin; a pumping part **200** that transfers the cosmetics, which are accommodated in the accommodation part **100**, to the head part **300**; and a cap part **500** that selectively covers the head part.

Hereinafter, detailed configurations of the cosmetic injector according to an embodiment of the present invention, on which microneedles for inducing absorption of cosmetics are mounted, will be described with reference to FIG. **3**.

First, the accommodation part **100** may include: an accommodation part body **110** for accommodating the cosmetics; and a bottom part **120** which is internally fitted to the accommodation part body **110** and maintains the level inside the accommodation part body **110** so that the cosmetics are discharged.

Next, the pumping part **200** may include: a fixed case **210** fixed to the accommodation part body **110**; a pumping case **220** which receives the cosmetics from the accommodation part body **110**, temporarily stores the cosmetics, and then guides the cosmetics to the head part **300**; a press part **230** that is lowered as the head part **300** is pressed; a spring **240** connected so that the press part **230** is elastically supported by the fixed case **210**; a push rod **250** which is raised or lowered by the press part **230** and transfers the cosmetics inside the pumping case **220** to the head part **300**; and a sealing cap **260** for forming a seal between the push rod **250** and the pumping case **220**.

The fixed case **210** may include: a fixed case body **211** having a cylindrical shape and fixed to the accommodation part body **110**; and a press part insertion **212** into which the press part is inserted and which has a coupling hole **213** that is a flow path through which the cosmetics move.

Also, the press part **230** may include: a press part body **231** supported by the spring **240**; and an insertion part **232** which is provided in the press part body **231**, has a discharge hole **233** serving as a path for moving the cosmetics, and is inserted into the head housing **310** and the coupling hole **213**.

The push rod **250** may include: a push rod body **251** having a cylindrical shape and a bore that serves as a path through which the cosmetics move; and an annular groove **252** formed in the outer circumference of the push rod body **251**.

The head part **300** may include: a first head part **320** for receiving the cosmetics from the pumping part **200**; a second head part **330** which is coupled to the upper portion of the first head part **320** and fixes the microneedle part **410** to the first head part **320**; and a head housing **310** for accommodating the first head part **320**, the second head part **330**, and the microneedle part **410**.

The head housing **310** may include: a head housing body **311** which has a cylindrical outer circumference and forms a resting part **311***a* for accommodating the first head part **320**, the second head part **330**, and the microneedle part **410**; a fixing protrusion **312** which protrudes upward from the middle wall of the head housing body **311** and fixes the positions of the first head part **320**, the second head part **330**, and the microneedle part **410**; and a head housing flow path **313** which is formed inside the head housing body **311** and guides the cosmetics to the first head part **320**.

Hereinafter, detailed configurations of the accommodation part **100** will be described with reference to FIG. **4**.

An accommodation part body **110** of the accommodation part **100** may include: a main body **111** having a cylindrical shape and accommodating the cosmetics; a partition wall **112** for preventing the bottom part **120** from separating from the lower portion of the accommodation part body **110**; and a partition wall ventilation opening **113** which is formed in the partition wall **112** and allows the bottom part **120** to communicate with the atmosphere and the main body **111** so that the bottom part **120** is lifted.

The bottom part **120** has an approximately cup shape, and is inserted into the main body **111** and brought into contact with the inner circumferential surface of the main body **111**.

Also, the cosmetics are accommodated between the pumping part **200** and the bottom part **120**, and the bottom part **120** may be moved toward the pumping part **200** by atmospheric pressure when the cosmetics are discharged to the outside of the main body **111** by the pumping part **200**.

Hereinafter, a structure for discharging the cosmetics from the accommodation part **100** to the head part **300** and a coupling relationship thereof will be described with reference to FIG. **5**.

The pumping case **220** may include: a pumping case body **221** which is coupled to the fixed case **210** and in which the cosmetics temporarily stays; and an inflow hole **222** which is provided in the lower portion of the pumping case body **221** and through which the cosmetics in the main body **111** flow in.

Also, the cosmetic injector according to an embodiment of the present invention, on which microneedles for inducing absorption of cosmetics are mounted, may further include a check valve **270** which is coupled to the inner lower portion of the pumping case body **221** and selectively opens and closes the inflow hole **222**.

The check valve **270** may open the inflow hole **222** due to negative pressure when the cosmetics accommodated in the main body **111** flow in the pumping case body **221**, and then may block the inflow hole **222** to prevent the cosmetics, which has flowed in the pumping case body **221**, from being discharged again to the main body **111**.

The insertion part **232** of the press part **230** is positioned above the pumping case **220** and inserted into the coupling hole **213**, and may move in the up-down direction along the coupling hole **213**.

The discharge hole **233** formed inside the insertion part **232** of the press part **230** communicates with the pumping case **220** via the push rod **250**.

Also, the spring **240** is located between the press part body **31** of the press part **230** and the fixed case body **211** and may elastically support the press part body **31** upward.

The push rod **250** is coupled to the press part **230**, and a lower portion thereof may be located inside the pumping case body **221**.

A press protrusion **234** of the press part **230** may be fastened to the annular groove **252** of the push rod **250**.

Also, a pumping hole **253** is formed in a lower portion of the push rod body **251** located inside the pumping case body **221**, and the pumping hole **253** may be opened and closed by the sealing cap **260**.

That is, the sealing cap **260** is formed surrounding the lower outer circumferential surface of the push rod **250**, and as described above, may open the pumping hole **253** when the push rod **250** descends and close the pumping hole **253** again when ascending.

Specifically, when the press part **230** and the push rod **250** are lowered, a space between the push rod **250** and the sealing cap **260** is open, and the cosmetics accommodated in the pumping case body **221** flow into the bore of the push rod body **251** via the pumping hole **253**. When the press part **230** and the push rod **250** are raised, the pumping hole **253** may be closed by the sealing cap **260**.

The head housing **310** may further include a resting groove **314** which is formed in the lower portion of the head housing body **311** and into which the insertion part **232** is inserted, and the cosmetics, which have flowed in from the bore of the push rod body **251**, may be guided to the head housing flow path **313** via the resting groove **314**.

FIG. **6** illustrates the lower surface of the first head part **320**.

The first head part **320** may include: a guide hole **324** that vertically passes through the first head part body **321** and guides the cosmetics to the second head part; and a first head part flow path **324** that is recessed from the lower portion of the first head part body **321** and formed radially about the through-hole.

Accordingly, the cosmetics, which have flowed into the head housing **310** via the pumping part, may uniformly spread on the first head part flow path **324** having the radial shape and then flow to the guide hole **324**.

That is, the first head part flow path **324** has the radial shape and temporarily stores the cosmetics and may serve to raise the pressure of the cosmetics that are discharged to the guide hole **324**.

Hereinafter, detailed structures of the first head part **320** and the second head part **330** will be described with reference to FIG. **7**.

The first head part **320** may include the first head part body **321** having a disc shape and a resting protrusion **322** protruding upward from the first head part body **321**. The resting protrusion **322** may be provided in plurality.

Also, the second head part **330** may include a second head part body **331** coupled to the first head part body **321** and a resting protrusion coupling part **332** serving as a hole into which the resting protrusion **322** and the plurality of microneedles **412** are inserted. The resting protrusion coupling part **332** may be provided in plurality.

The detailed structure of the microneedle part **410** and the coupling relationship between the first head part **320** and the second head part **330** will be described with reference to FIG. **8**.

The microneedle part **410** may include: a microneedle plate **411** coupled between the first head part body **321** and the second head part body **331**; and a plurality of microneedles **412** which extend from the microneedle plate **411**, protrude toward the upper surface of the second head part **330**, and guide the cosmetics, discharged from the head part, into the skin.

Also, a resting protrusion through-hole **413**, into which the resting protrusion **322** is inserted, may be formed in the microneedle plate **411**. The plurality of microneedles **412** extend laterally from the resting protrusion through-hole **413** and the edge of the microneedle plate **411** and then bent, and thus may be perpendicular to the microneedle plate **411**.

The microneedle part **410** may be made of metal so that contact points between the microneedle plate **411** and the plurality of microneedles **412** can be pressed and bent by the resting protrusion, and the microneedle plate **411** and the plurality of microneedles **412** may be integrally formed.

Accordingly, the present application can enhance coupling and durability of the microneedle part **410** and thus easily prevent breakage of microneedles that occurs in a cosmetic injector in which microneedles according to the related art are mounted.

In addition, the present application eliminates the need to manually install each of a plurality of microneedles in the head part, thereby reducing working costs and preventing each of the microneedles from being contaminated due to the manual work of an operator during the manufacturing process.

Meanwhile, the first head part **320** may further include a first head part fixing protrusion insertion hole (not shown) through which the fixing protrusion **312** of the head housing **310** passes, the microneedle part **410** may further include a microneedle part fixing protrusion insertion hole **414** through which the fixing protrusion **312**, which has passed through the first head part fixing protrusion insertion hole, passes, and the second head part **330** may further include a second head part fixing protrusion insertion hole **333** through which the fixing protrusion **312**, which has passed through the microneedle part fixing protrusion insertion hole **413**, passes.

Accordingly, the microneedle part **410** may rest at the right position between the first head part **320** and the second head part **330**, and thus assemblability may be improved.

Here, the fixing protrusion **312**, the first head part fixing protrusion insertion hole, the microneedle part fixing protrusion insertion hole **414**, and the second head part fixing protrusion insertion hole **333** may be provided in plurality.

FIG. **9** illustrates a process in which the microneedle part **410** is bent when a product is assembled.

First, the plurality of microneedles **412** extend toward the edge of the microneedle plate **411** and the resting protrusion through-hole **413** and formed horizontally with the microneedle plate **411**.

Subsequently, the plurality of microneedles **412** are pressed upward by the resting protrusion **322** while resting on the first head part **320**, and thus bent in a direction perpendicular to the microneedle plate **411**.

That is, when the microneedle plate **411** is pressed downward against the first head part **320** by another mechanism, on which the microneedle plate **411** rests, or the second head part **330**, the plurality of microneedles **412** may be simultaneously bent upward and then inserted into the resting protrusion coupling part **332**, and thus may be positioned between the resting protrusion **322** and the second head part body **331**.

Subsequently, the vertically bent microneedles **412** pass through gaps (see FIG. **10**) formed between the resting protrusion **322** and the resting protrusion coupling part **332** and protrude above the second head part **330**.

When the gaps for accommodating microneedles are formed, the microneedle are fixed, and bending of the microneedles is prevented. Moreover, the cosmetics may be smoothly discharged to the outside of the head part via the through-holes formed between the microneedles.

Also, the length of each of the plurality of microneedles **412** may be greater than the distance from the upper surface of the first head part body **321** to the upper surface of the second head part body **331**. The length of the microneedle **412** protruding upward from the second head part may be preferably 0.001 to 2 mm.

When the length of a portion of the end of the microneedle **412**, which protrudes upward from the second head part **330**, is less than 0.001 mm, there is almost no skin penetration effect of cosmetics. When the length is greater than 2 mm, the capillaries may rupture, and bleeding may occur. Therefore, the protrusion length thereof is preferably limited to the above-mentioned range.

When the length of the bent microneedle is greater than the distance from the resting part to the upper surface of the head part, the microneedle may be allowed to protrude to the outside of the head part. Moreover, the microneedle is supported by the second head part, and thus durability thereof may be enhanced.

The microneedle part **410** may be preferably manufactured by cutting a plate, more preferably manufactured by cutting a metal plate, even more preferably manufactured by press or laser cutting a metal plate, or most preferably manufactured by press cutting a metal plate.

Next, the operation and use method of the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to an embodiment of the present invention will be described.

First, the cap part **500** is separated from the head part **300**, and the second head part **330** is brought into contact with the skin. Then, when the accommodation part body **110** is pressed toward the second head part **330**, the second head part **330** is brought into close contact with the skin. The microneedles **412** pierce the epidermis of the skin, and the tip ends are inserted below the epidermis.

During this process, the spring **240** is compressed as the press part **230** moves toward the pumping case **220**, and a gap is made between the push rod **250** and the sealing cap **260**. The cosmetics accommodated inside the pumping case **220** are allowed to flow into the push rod **250** via the pumping hole **253**. Here, the inflow hole **222** is blocked by the check valve **270**.

The cosmetics, which have flowed into the push rod **250** through this process, are transferred to the first head part flow path **323** via the discharge hole **233** and the head housing flow path **313**.

Subsequently, the cosmetics flow to the central portion along the first head part flow path **323** and are then transferred to the second head part **330** via the guide hole **324**. The cosmetics arriving at the second head part **330** may be discharged to the outside of the second head part via the gaps formed between the resting protrusion **322** and the resting protrusion coupling part **332**.

Subsequently, when the pressure applied on the accommodation part body **110** is released, the press part **230** is returned to the original position by the elastic force of the spring **240**. At the same time, the push rod **250** and the sealing cap **260** move upward.

Here, the pumping hole **253** formed in the lower portion of the push rod **250** is blocked by the sealing cap **260**, and the inflow hole **222** is opened by the check valve **270** as the pressure inside the pumping case **220** becomes lower.

Accordingly, the cosmetics accommodated inside the accommodation part body **110** are allowed to flow into the pumping case body **221** via the inflow hole **222**, and the bottom part **120** moves upward, thereby always maintaining a state in which the space between the pumping part **200** and the bottom part **120** is completely filled with the cosmetics.

Next, a method for manufacturing the cosmetic injector, on which microneedles for inducing absorption of cosmetics are mounted, according to an embodiment of the present invention will be described.

The manufacturing method for the cosmetic injector may include: a resting operation (S1) of resting the microneedle plate on the upper surface of the first head part in a state in which the microneedle plate **411** and the plurality of microneedles **412** are provided parallel to each other; and a head part coupling operation (S2) of coupling the second head part **330** to the upper portion of the first head part **320**.

Here, the plurality of microneedles **412** may be bent perpendicularly to the microneedle plate **411** while the microneedle plate **411** rests on the first head part **320** and is then coupled thereto.

When the microneedles are bent perpendicularly while the microneedle plate rests on the resting part, it is possible to obtain the effect of reducing the manufacturing cost by omitting a separate press process for bending the microneedles.

In the present invention, each of the above-described components may additionally have another effect, even if the effect is not described herein. Also, it is possible to derive a new effect that cannot be found in the related art according to the organic coupling relationship between the above-described components.

In addition, embodiments illustrated in the drawings may be modified and implemented in other forms. These modifications should be regarded as falling within the scope of the present invention when the modifications are carried out so as to include components in the claims or within the scope of equivalents thereof.

The invention claimed is:

1. A cosmetic injector, on which a plurality of microneedles for inducing absorption of cosmetics are mounted, the cosmetic injector comprising:

an accommodation part configured to accommodate cosmetics;

a head part having a flow path through which the cosmetics flow in;

a microneedle part which is fixed to the head part, and which is insertable into the skin and allows the cosmetics to penetrate the skin; and

a pumping part configured to transfer, to the head part, the cosmetics accommodated in the accommodation part,

wherein the head part comprises:

a first head part configured to receive the cosmetics from the pumping part; and

a second head part coupled to an upper portion of the first head part,

wherein the microneedle part comprises:

a microneedle plate provided between the first head part and the second head part; and

the plurality of microneedles which extend from the microneedle plate and protrude toward an upper surface of the second head part, and which guide, into the skin, the cosmetics discharged from the head part,

wherein the plurality of microneedles is formed integrally and horizontally with the microneedle plate,

wherein the first head part comprises:

a first head part body having an upper surface on which the microneedle plate rests; and

a resting protrusion having an upper surface portion horizontal to the microneedle plate and the upper surface of the first head part body, and a sidewall portion vertical to the upper surface portion, wherein the upper surface portion contacts the plurality of microneedles,

wherein the microneedle plate has a resting protrusion through-hole through which the resting protrusion passes, and

wherein the microneedle plate is made of metal so that, when the resting protrusion passes through the resting protrusion through-hole, the plurality of microneedles is pressed by the upper surface portion of the resting protrusion and bent in a vertical direction along the sidewall portion of the resting protrusion.

2. The cosmetic injector of claim **1**, wherein the resting protrusion is provided in plurality.

3. The cosmetic injector of claim **1**, wherein the plurality of microneedles extend upward from a vicinity of the resting protrusion through-hole of the microneedle plate or from edges of the microneedle plate.

4. The cosmetic injector of claim **1**, wherein the second head part comprises a second head part body having a lower surface that contacts the microneedle plate, and

the second head part body has a resting protrusion coupling part that is a hole into which the resting protrusion and the plurality of microneedles are inserted.

5. The cosmetic injector of claim 4, wherein gaps configured to accommodate the plurality of microneedles are formed between the resting protrusion coupling part and the resting protrusion or between the outer boundary of the second head part body and the resting protrusion.

6. The cosmetic injector of claim 5, wherein the length of each of the plurality of microneedles is greater than the distance from the upper surface of the first head part body to the upper surface of the second head part.

7. The cosmetic injector of claim 1, further comprising a head housing that accommodates the first head part, the second head part, and the microneedle part, and receives the cosmetics from the pumping part.

8. The cosmetic injector of claim 7, wherein the first head part body has:

a guide hole that vertically passes through the first head part body; and

a first head part flow path which is provided in a lower portion of the first head part body and formed radially about the guide hole,

wherein the cosmetics, which have flowed into the head housing via the pumping part, are guided to the guide hole via the first head part flow path.

9. The cosmetic injector of claim 7, wherein the second head part comprises a second head part body having a lower surface that contacts the microneedle plate,

wherein the head housing further comprises:

a head housing body configured to accommodate the first head part, the second head part, and the microneedle part; and

a fixing protrusion that protrudes from the head housing body,

wherein the first head part further comprises a first head part fixing protrusion insertion hole through which the fixing protrusion passes,

the microneedle part further comprises a microneedle part fixing protrusion insertion hole through which the fixing protrusion, which has passed through the first head part fixing protrusion insertion hole, passes, and

the second head part further comprises a second head part fixing protrusion insertion hole through which the fixing protrusion, which has passed through the microneedle part fixing protrusion insertion hole, passes.

10. The cosmetic injector of claim 9, wherein the fixing protrusion is provided in plurality, and

the number of each of first head part fixing protrusion insertion holes, microneedle part fixing protrusion insertion holes, and second head part fixing protrusion insertion holes corresponds to the number of fixing protrusions.

11. The cosmetic injector of claim 1, wherein when the head part is pressed toward the pumping part, the cosmetics, which have been supplied via the pumping part, are discharged via the outer circumferential surfaces of the plurality of microneedles.

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